# Cross-border interconnectors Why they exist

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Cross-border interconnectors - why they exist and contribute to a sustainable energy future

Cross-border interconnectors play an important role in maintaining security of power supply. Additionally, they allow us to utilize our power as efficiently as possible by selling it to neighboring countries. Cooperation ensures that resources are used more effectively.

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Previously, it was a local responsibility to ensure that residents had enough electricity. One had to account for and build enough production capacity for dry years to avoid power shortages and rationing. This is not very beneficial for society because building such overcapacity incurs high costs, which in turn makes electricity more expensive. Additionally, it comes at the expense of nature and the environment.

Gradually, power systems were interconnected. This cooperation allows us to utilize resources much better, strengthening the power system and supply security. This has helped keep electricity costs down. For the same reason, the connection to Sweden in the 1960s marked the beginning of cooperation with our neighboring countries.

Cross-Border Interconnectors Allow Us to Benefit from Other Energy Sources

Norway's electricity production largely comes from hydropower. This means that electricity production and prices are closely linked to precipitation throughout the year and the amount of water in the reservoirs. This makes us vulnerable during periods of low precipitation. With cross-border interconnectors, we can buy electricity as a nation during periods when we need it, ensuring that we meet our energy needs.

To achieve Europe's goal of reducing emissions, production and consumption are shifting from fossil to renewable energy. The need for more power will also increase significantly in the coming years. The challenges we face now are in many ways comparable to the development of the Norwegian power system. To meet demand, all countries must improve efficiency and expand power production, but power exchange with our neighboring countries ensures that we do not build more power or reduce more nature than necessary.

Cross-border interconnectors also help us optimize the operation and use of water at home.

A Weather-Dependent Power System Increases the Need for Cooperation

Europe is moving towards a renewable future where weather-dependent power production will play a much larger role, leading to more variable production. In Norway, we have an additional advantage with flexible hydropower that can be relatively easily regulated up and down.

Our neighboring countries have production that cannot, or is not as profitable to, regulate as quickly. When consumption decreases, this power production continues. We can then quickly reduce much of the hydropower production and import cheaper surplus power from other countries.

The same effect occurs when it is windy or sunny. Then production is high relative to demand, prices go down, and we can import instead of using our own water. When consumption increases, we can increase hydropower production and export when demand is highest. To solve the challenges of a weather-dependent power system, we are entirely dependent on an interconnected power market.

Cross-border interconnectors make us more affected by both high and low electricity prices in the surrounding countries. Cooperation and exchange are important for both Norway and Europe to have continuous access to power and contribute to the efficient use of resources.